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Acknowledgements

AI tools such as Claude, ChatGPT was used to support:

- Drafting and refining documentation (Developer Guide and User Guide)
- Generating and improving test case structures
- Suggesting code refactoring and design clarifications
- All AI-generated content was reviewed, validated, and adapted by the development team to ensure correctness and alignment with project requirements.

Setting up, getting started

Refer to *Setting up and getting started*.

Design

:bulb: **Tip:** PlantUML source files are in `docs/diagrams`. This guide assumes rendered diagram images are also available under `docs/diagrams`.

Architecture

Architecture Diagram

> **UML Placeholder**: Replace `diagrams/ArchitectureDiagram.png` with the rendered output of `docs/diagrams/ArchitectureDiagram.puml`.

CrimeWatch follows a layered architecture with five main parts:

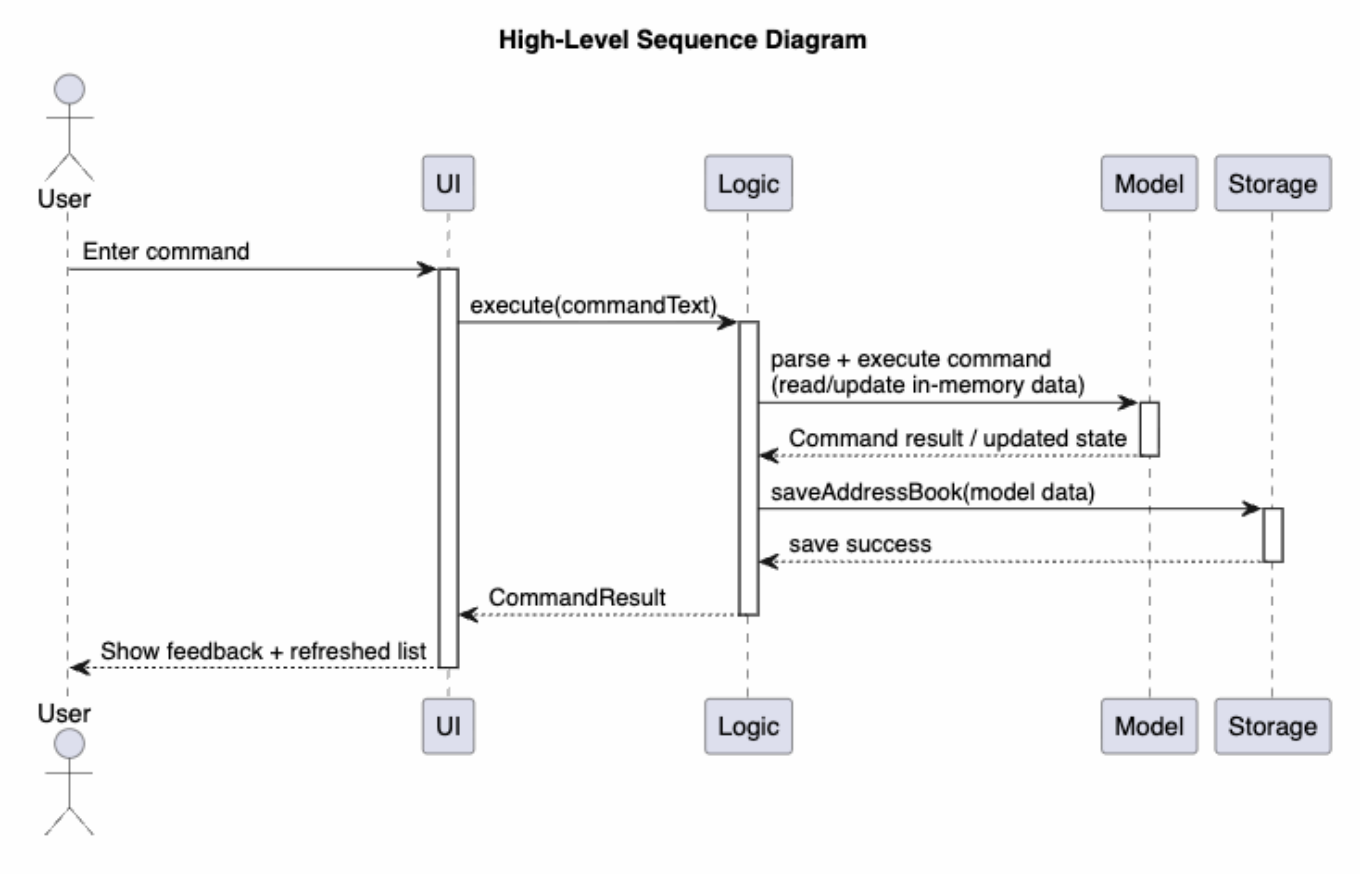
- `UI`: renders data and captures user input.
- `Logic`: parses and executes commands.
- `Model`: stores in-memory domain state.
- `Storage`: persists and loads data.
- `Commons`: shared helpers (`logs`, `config`, utility classes).

`MainApp` initializes `Storage`, `Model`, `Logic`, and `UI` in this order, and wires dependencies through interfaces.

At runtime, a typical command follows this high-level flow:

1. User enters a command in `UI`.
2. `Logic` parses the input into a concrete `Command`.
3. `Command` updates or queries `Model`.
4. `LogicManager` triggers persistence through `Storage`.
5. `UI` refreshes based on observable lists and command result feedback.

High-level runtime sequence



The diagram above summarizes the end-to-end interaction across `UI`, `Logic`, `Model`, and `Storage` for a typical command execution.

Architecture Sequence Diagram

> **UML Placeholder**: Replace `diagrams/ArchitectureSequenceDiagram.png` with the rendered output of `docs/diagrams/ArchitectureSequenceDiagram.puml`.

Component Managers Diagram

> **UML Placeholder**: Replace `diagrams/ComponentManagers.png` with the rendered output of `docs/diagrams/ComponentManagers.puml`.

UI component

API

`Ui.java`

UI Class Diagram

> **UML Placeholder**: Replace `diagrams/UiClassDiagram.png` with the rendered output of `docs/diagrams/UiClassDiagram.puml`.

The UI is JavaFX-based and centered around `MainWindow`, which composes:

- `CommandBox` for command input.
- `ResultDisplay` for feedback messages.

- `PersonListPanel` for list rendering.
- `ViewPanel` for full contact profile details.
- `StatusBarFooter` for save path/status.
- `HelpWindow` for help content.

`UiManager` owns startup concerns (main stage setup, icon setup, fatal dialog handling), while `MainWindow` handles most user interaction wiring.

Design notes:

- `MainWindow` receives a `Logic` reference and executes commands through `logic.execute(...)`.
- `PersonListPanel` is bound to `logic.getFilteredPersonList()`.
- `CommandResult` flags (`showHelp`, `exit`, `personToView`) drive post-command UI behavior.

Logic component

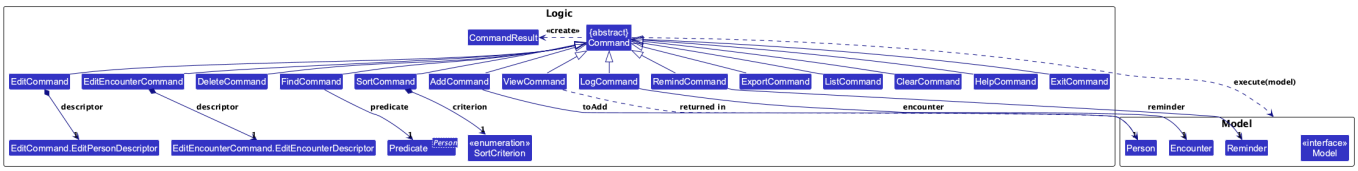
API

`Logic.java`

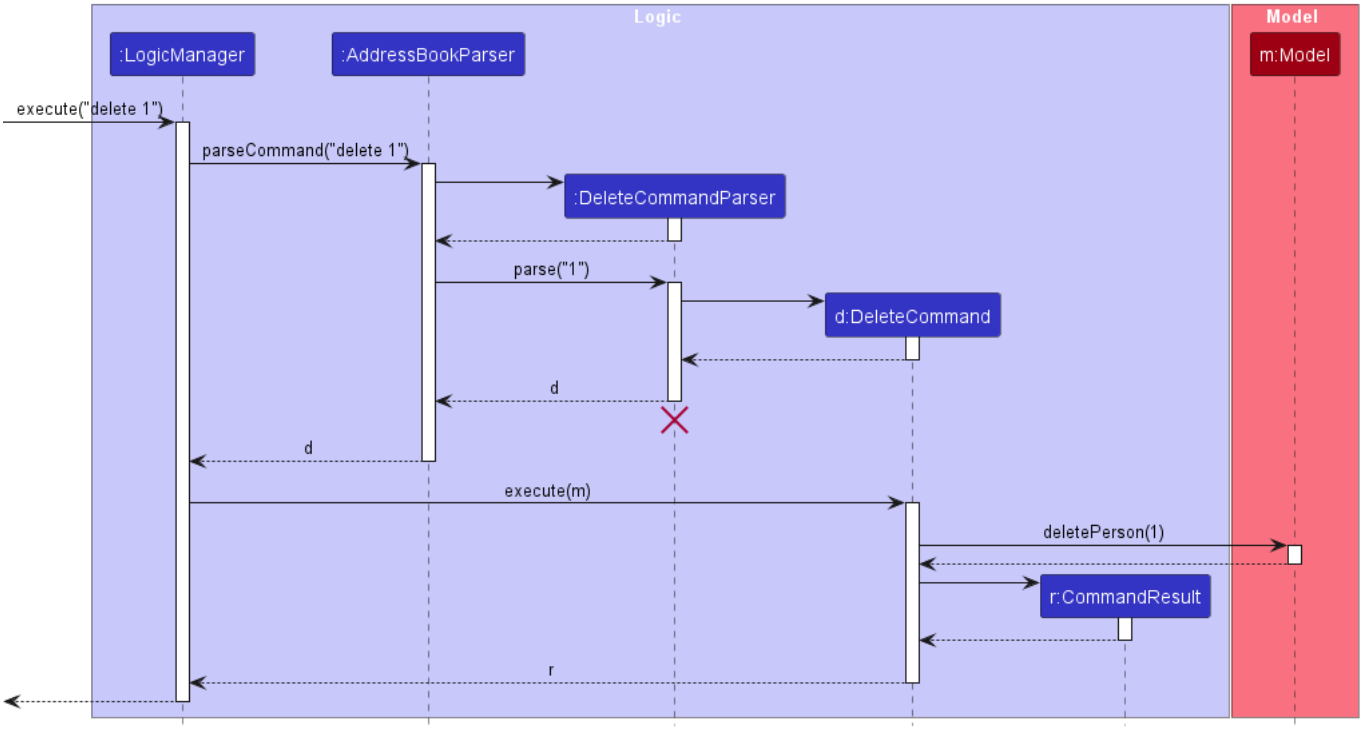
Logic Class Diagram

> **UML Placeholder**: Replace `diagrams/LogicClassDiagram.png` with the rendered output of `docs/diagrams/LogicClassDiagram.puml`.

The diagram above treats concrete command types as `XYZCommand` to keep the high-level view compact. The following class diagram zooms into that placeholder and shows the concrete command hierarchy, together with key command-specific collaborators and multiplicities:



The sequence diagram below illustrates the interactions within the `Logic` component, taking `execute("delete 1")` API call as an example.



:information_source: **Note:** The lifeline for `DeleteCommandParser` should end at the destroy marker (X) but due to a limitation of PlantUML, the lifeline continues till the end of diagram.

The `Logic` component processes command text in this pipeline:

1. `LogicManager.execute(String)` receives raw input.
2. `AddressBookParser` determines command word and delegates to command-specific parser.
3. A concrete `Command` object is created and executed on `Model`.
4. `CommandResult` is returned to `UI`.
5. `LogicManager` persists state by calling `storage.saveAddressBook(...)`.

Parser Classes Diagram

> **UML Placeholder**: Replace `diagrams/ParserClasses.png` with the rendered output of `docs/diagrams/ParserClasses.puml`.

Parser design:

- Command parsers implement the common `Parser<T>` interface.
- Prefix-based commands use `ArgumentTokenizer` and `ArgumentMultimap`.
- Validation and conversion are centralized in `ParserUtil`.
- Duplicate-prefix validation is handled by `verifyNoDuplicatePrefixesFor(...)`.

Model component

API

`Model.java`

Model Class Diagram

> **UML Placeholder**: Replace `diagrams/ModelClassDiagram.png` with the rendered output of `docs/diagrams/ModelClassDiagram.puml`.

`ModelManager` stores in-memory app state:

- `AddressBook` for canonical contact data.
- `UserPrefs` for application preferences.
- `FilteredList<Person>` + `SortedList<Person>` for UI-facing views.

Core domain type: `Person`, extended for CrimeWatch with:

- identity/contact fields (`Name`, `Phone`, `Email`, `Address`)
- investigation fields (`Stage`, `Risk`, `Alias`, `Notes`)
- collections (`Tag`, `Encounter`, `Reminder`)
- optional `Password` for feature-level access control

Design notes:

- identity uniqueness is enforced through `UniquePersonList` and `Person#isSamePerson`.
- sorting is view-level (via comparator on `SortedList`) and does not reorder persisted storage sequence.
- reminder entries are maintained in chronological order when added.

Storage component

API

`Storage.java`

Storage Class Diagram

> **UML Placeholder**: Replace `diagrams/StorageClassDiagram.png` with the rendered output of `docs/diagrams/StorageClassDiagram.puml`.

Storage uses JSON persistence:

- `StorageManager` coordinates address book and preferences storage.
- `JsonAddressBookStorage` reads/writes contact data.
- `JsonUserPrefsStorage` reads/writes GUI/user preferences.
- `JsonSerializableAddressBook` + `JsonAdaptedPerson` bridge model objects and JSON schema.

Error behavior:

- Read/load errors surface to startup logic.
- Save errors during command execution are wrapped as user-visible `CommandException`s in `LogicManager`.

Common classes

The `seedu.address.commons` package contains shared utilities:

- `LogsCenter` and logger setup.
- `GuiSettings` and config helpers.
- value/helper utilities (`StringUtil`, `ToStringBuilder`, etc.).

Implementation

This section documents key implementation decisions for major CrimeWatch features.

Add command

Overview

`add` creates a new `Person` with required and optional fields:

- required: `n/`, `p/`, `e/`, `a/`, `s/`
- optional: `al/`, `note/`, `r/`, `pw/`, `t/`

Command format

The current `add` command format is:

```
add n/NAME p/PHONE e/EMAIL a/ADDRESS s/STAGE [al/ALIAS(,ALIAS...)] [note/NOTES]
[r/RISK] [pw/PASSWORD] [t/TAG]...
```

Required fields:

- `n/` name
- `p/` phone
- `e/` email
- `a/` address
- `s/` stage

Optional fields:

- `al/` aliases (comma-separated)
- `note/` notes
- `r/` risk (defaults to `medium`)
- `pw/` password (contact-level protection)
- `t/` tags (repeatable)

Parsing flow

`AddCommandParser`:

- tokenizes input by supported prefixes.
- validates required prefixes and preamble constraints.
- rejects duplicate single-value prefixes.
- parses fields through `ParserUtil` and constructs `Person`.

`AddCommand#execute(Model)`:

- checks duplicate identity via `model.hasPerson(toAdd)`.
- inserts via `model.addPerson(toAdd)`.

Key parser behavior:

- Rejects missing required prefixes with `MESSAGE_INVALID_COMMAND_FORMAT`.
- Uses `verifyNoDuplicatePrefixesFor(...)` for `n/`, `p/`, `e/`, `a/`, `al/`, `note/`, `r/`, `pw/`, `s/`.
- Parses aliases via `ParserUtil.parseAliases(...)`; empty alias payload is rejected.
- Applies default risk via `Risk.getDefault()` when `r/` is omitted.
- Parses tags from all `t/` occurrences into a `Set<Tag>`.

Field constraints

Validation is enforced in model/value objects and parser utilities:

- `Name`: alphanumeric + spaces, non-blank.
- `Phone`: digits only, at least 3 digits.
- `Email`: must satisfy email format constraints.
- `Address`: non-blank.
- `Alias`: trimmed, 1-50 chars, alphanumeric + spaces.
- `Stage`: one of `surveillance`, `approached`, `cooperating`, `arrested`, `closed`.
- `Notes`: optional text, max 500 chars, no newlines.
- `Risk`: one of `low`, `medium`, `high` (case-insensitive parser).
- `Password`: alphanumeric + spaces, non-blank when provided.
- `Tag`: alphanumeric.

Relevant classes:

- `src/main/java/seedu/address/logic/parser/ParserUtil.java`
- `src/main/java/seedu/address/model/person/Name.java`
- `src/main/java/seedu/address/model/person/Phone.java`
- `src/main/java/seedu/address/model/person/Email.java`
- `src/main/java/seedu/address/model/person/Address.java`
- `src/main/java/seedu/address/model/person/Alias.java`
- `src/main/java/seedu/address/model/person/Stage.java`
- `src/main/java/seedu/address/model/person/Notes.java`
- `src/main/java/seedu/address/model/person/Risk.java`
- `src/main/java/seedu/address/model/person/Password.java`
- `src/main/java/seedu/address/model/tag/Tag.java`

Sort feature

Overview

The `sort` feature reorders the currently displayed contact list by a selected criterion:

- `location`
- `tag`

- `alphabetical`
- `status`
- `recent`

This is implemented as a **view-level sort**. It does not mutate persisted `AddressBook` ordering in storage.

Command flow

1. User enters `sort CRITERION`.
2. `AddressBookParser` routes the command word `sort` to `SortCommandParser`.
3. `SortCommandParser` validates that there is exactly one token and maps it to `SortCriterion`.
4. `SortCommand#execute(Model)` calls `model.setPersonSortComparator(...)` with the criterion comparator.
5. UI updates automatically because it is bound to `Model#getFilteredPersonList()`.

Key classes:

- `src/main/java/seedu/address/logic/parser/AddressBookParser.java`
- `src/main/java/seedu/address/logic/parser/SortCommandParser.java`
- `src/main/java/seedu/address/logic/commands/SortCommand.java`

Model integration

Sorting support is added to the `Model` API:

- `setPersonSortComparator(Comparator<Person>)`
- `clearPersonSortComparator()`

`ModelManager` now keeps:

- `FilteredList<Person> filteredPersons` for filtering
- `SortedList<Person> sortedPersons` wrapping `filteredPersons` for sorting

`getFilteredPersonList()` returns `sortedPersons`, so existing UI wiring works without extra UI changes.

Key classes:

- `src/main/java/seedu/address/model/Model.java`
- `src/main/java/seedu/address/model/ModelManager.java`

Comparator behavior

Implemented in `SortCommand.SortCriterion`:

- `alphabetical`: by `Person#getName()`
- `status`: by `Person#getStage().toString()`, then by name
- `tag`: by alphabetically smallest tag name, nulls last, then by name
- `location`: by latest encounter location, normalized (trim + collapse spaces), nulls last, then by name
- `recent`: by latest encounter datetime descending (most recent first), then by name

For contacts with missing values (e.g., no encounters/no tags), comparators use `nullsLast(...)` so they appear at the end for those criteria.

Defensive parsing and validation

`SortCommandParser` rejects:

- missing criterion
- multiple tokens
- unsupported criterion

All invalid forms throw `ParseException` with `MESSAGE_INVALID_COMMAND_FORMAT` and command usage.

Tests

Parser tests:

- `src/test/java/seedu/address/logic/parser/SortCommandParserTest.java`
- registration coverage in `src/test/java/seedu/address/logic/parser/AddressBookParserTest.java`

Command/model integration tests:

- `src/test/java/seedu/address/logic/commands/SortCommandTest.java`

Compatibility updates:

- `Model` test stubs implement the new methods, e.g. in `src/test/java/seedu/address/logic/commands/AddCommandTest.java`

Log command

The `log` command appends a new encounter to the selected contact in the current displayed list. After validating the target index against `Model#getFilteredPersonList()`, `LogCommand#execute(Model)` copies the contact's existing encounters, appends the new `Encounter`, rebuilds the `Person`, and calls `model.setPerson(...)`.

The reconstructed `Person` preserves all unrelated fields, including reminders and password, while updating only the encounter history. The returned `CommandResult` includes the updated person so the UI can continue showing the refreshed profile.

Key classes:

- `src/main/java/seedu/address/logic/commands/LogCommand.java`
- `src/main/java/seedu/address/model/person/Encounter.java`
- `src/main/java/seedu/address/model/person/Person.java`

Export encounters to CSV

Overview

`export l/LOCATION` writes encounters at the specified location to: `exports/CrimeWatch-export-<timestamp>.csv`.

Implementation notes

- `ExportCommandParser` validates the required `l/LOCATION` input.
- `ExportCommand` scans the canonical address book via `model.getAddressBook().getPersonList()`, so export is independent of the current filtered or sorted UI view.
- Matching is case-insensitive after trimming surrounding whitespace.
- Matching encounters are converted into CSV rows, sorted by encounter datetime, and written to `./exports/`.
- The output directory is created automatically if it does not already exist.

Rationale and trade-off

- CSV is easy to inspect and import into reporting tools.
- Exact location matching keeps export behavior predictable, while case-insensitive comparison remains forgiving for user input.

Password Feature

Overview

Optional, contact-level password protection. Each contact can be protected with a password to restrict viewing its full details.

Feature	Description
Scope	Per-contact (individual contacts can be protected)
Type	Optional (contacts don't require passwords)
Usage	Add `pw/PASSWORD` to `add` or `edit` commands to protect; provide it with `view` to access
Validation	Alphanumeric characters and spaces only

Usage

```
# Add contact with password protection
add n/John Doe p/98765432 e/john@example.com a/123 Main St s/surveillance
pw/password123

# Update/remove password
edit 1 pw/newpassword   # Change password
edit 1 pw/              # Remove protection
```

View protected contact

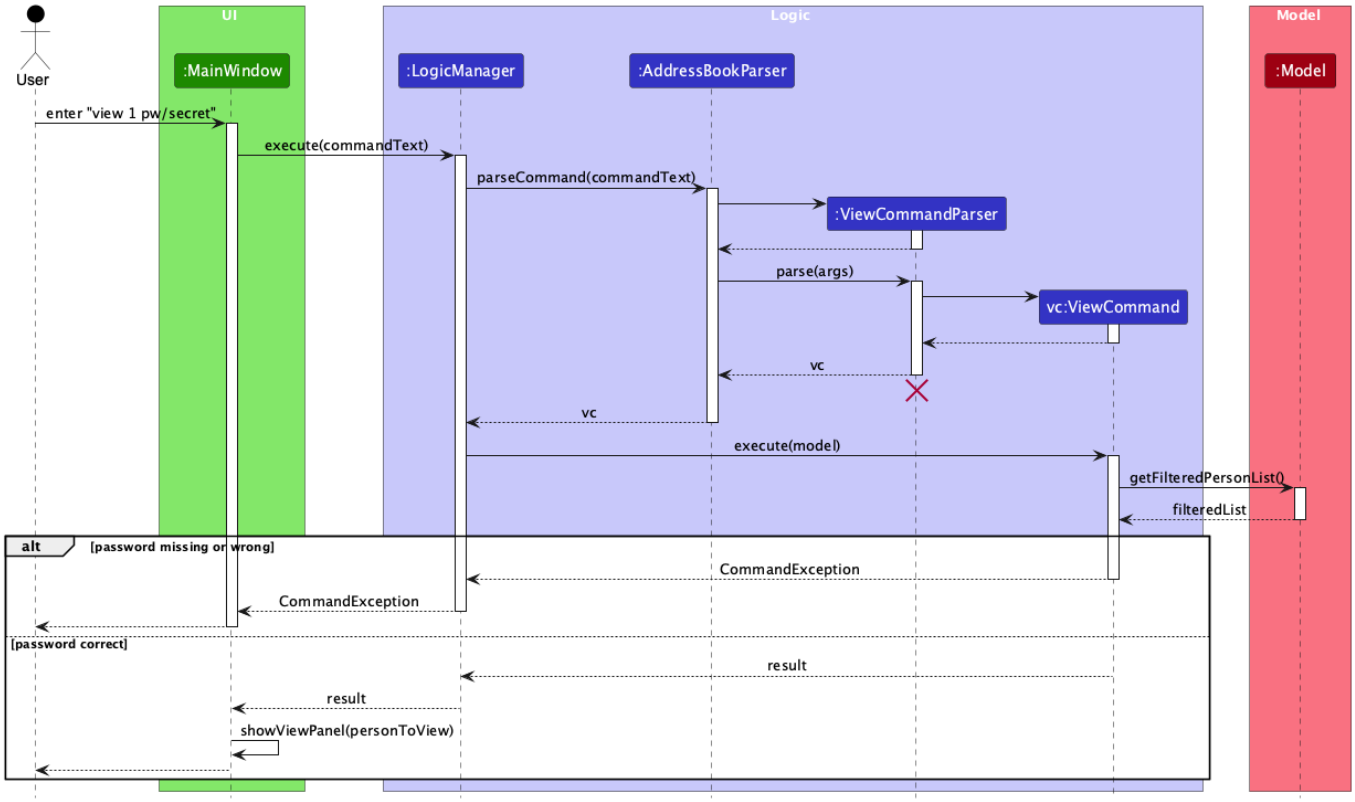
view 1 pw/password123 # Show full details if password correct view 1 # Error: password required

Behavior

- **Without password**: Contact viewable normally
- **With password**: `view` command requires correct password to display full details
- **Plain text**: Passwords stored without encryption (not production-ready)

Sequence Diagram

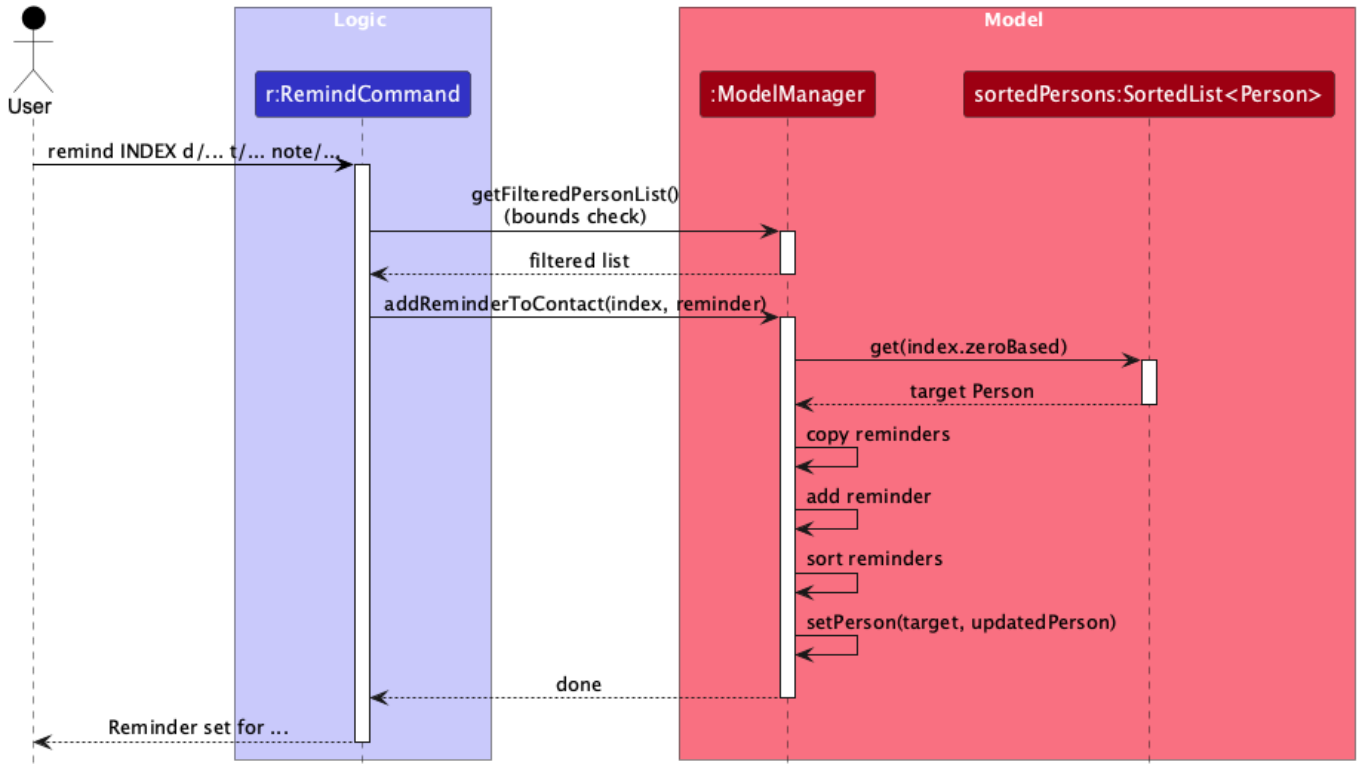
The following sequence diagram shows how the `view` command processes a password-protected contact:



Remind command

The `remind` command resolves its target using the same filtered and sorted contact list currently shown to the user. This is important because `ModelManager#getFilteredPersonList()` returns `sortedPersons`, so the displayed index must be interpreted against the sorted view rather than the raw address book order.

The command first performs bounds checking against the displayed list in `RemindCommand#execute(Model)`, then delegates the actual update to `Model#addReminderToContact(Index, Reminder)`. `ModelManager` retrieves the target from `sortedPersons`, copies the existing reminder list, adds the new reminder, sorts the updated reminders chronologically, and rebuilds the `Person` before calling `setPerson(...)`.



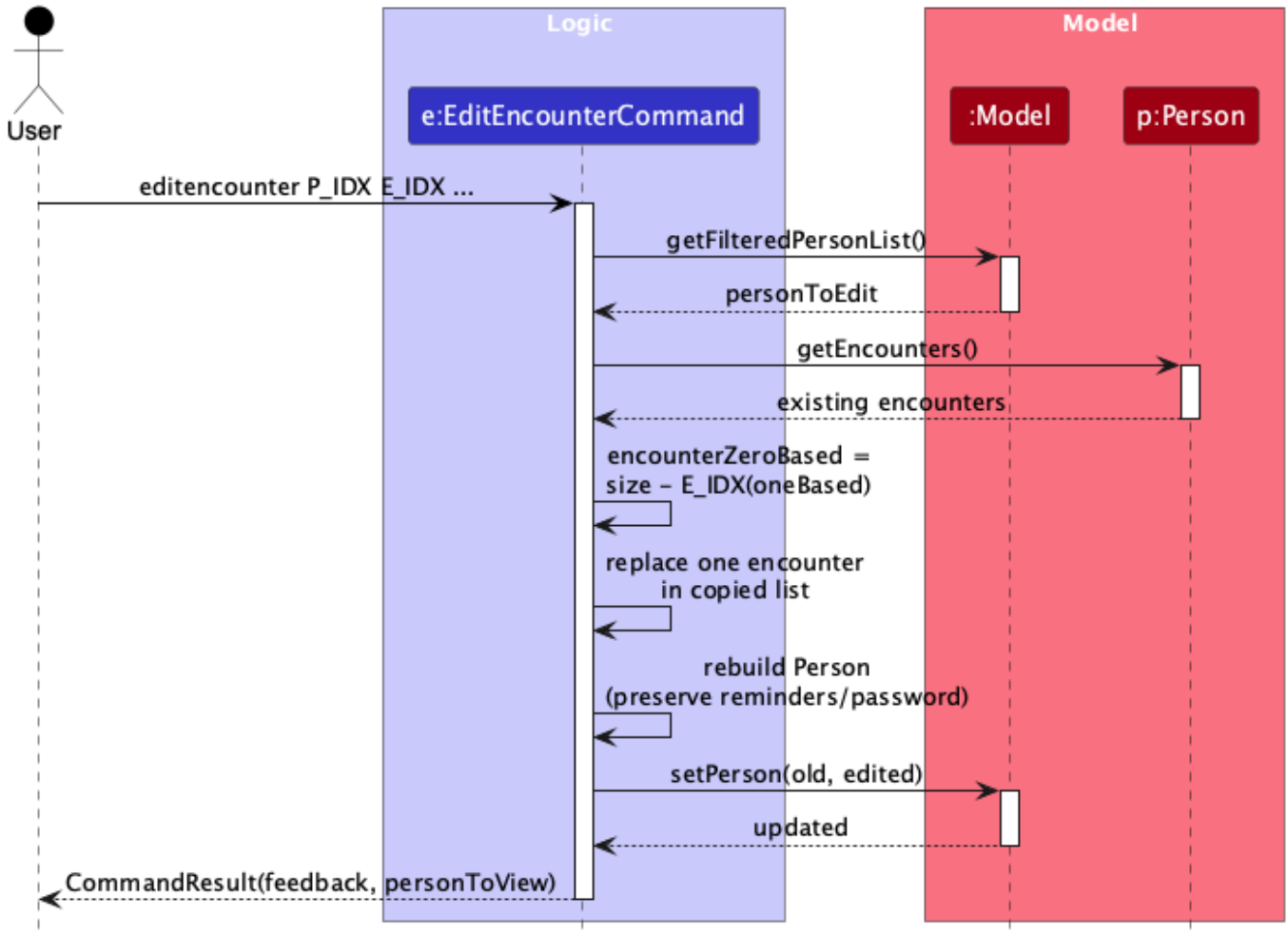
Key classes:

- `src/main/java/seedu/address/logic/commands/RemindCommand.java`
- `src/main/java/seedu/address/model/Model.java`
- `src/main/java/seedu/address/model/ModelManager.java`

Edit Encounter command

The `editencounter` command uses two indices: `PERSON_INDEX` identifies the contact in the displayed contact list, while `ENCOUNTER_INDEX` refers to the encounter cards shown in `view`. The UI renders those encounter cards in reverse-chronological order, so display index `1` refers to the most recent encounter rather than the first element stored in the underlying encounter list.

To preserve that user-facing numbering, `EditEncounterCommand#execute(Model)` converts the displayed encounter index back into the stored zero-based index using `existingEncounters.size() - encounterDisplayOneBased`. It then replaces only the selected encounter in a copied list and rebuilds the `Person` with the updated encounters while preserving other fields such as reminders and password protection.



Key classes:

- `src/main/java/seedu/address/logic/commands/EditEncounterCommand.java`
- `src/main/java/seedu/address/ui/ViewPanel.java`
- `src/main/java/seedu/address/model/person/Person.java`

[Proposed] Undo/redo feature

Proposed Implementation

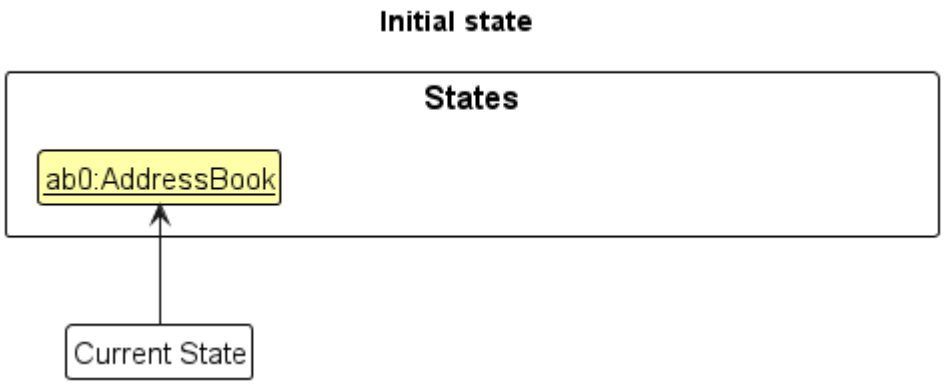
The proposed undo/redo mechanism is facilitated by `VersionedAddressBook`. It extends `AddressBook` with an undo/redo history, stored internally as an `addressBookStateList` and `currentStatePointer`. Additionally, it implements the following operations:

- `VersionedAddressBook#commit()` — Saves the current address book state in its history.
- `VersionedAddressBook#undo()` — Restores the previous address book state from its history.
- `VersionedAddressBook#redo()` — Restores a previously undone address book state from its history.

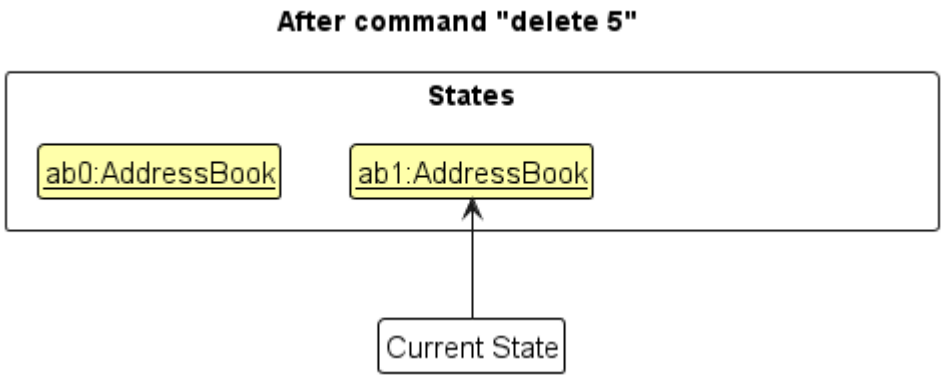
These operations are exposed in the `Model` interface as `Model#commitAddressBook()`, `Model#undoAddressBook()` and `Model#redoAddressBook()` respectively.

Given below is an example usage scenario and how the undo/redo mechanism behaves at each step.

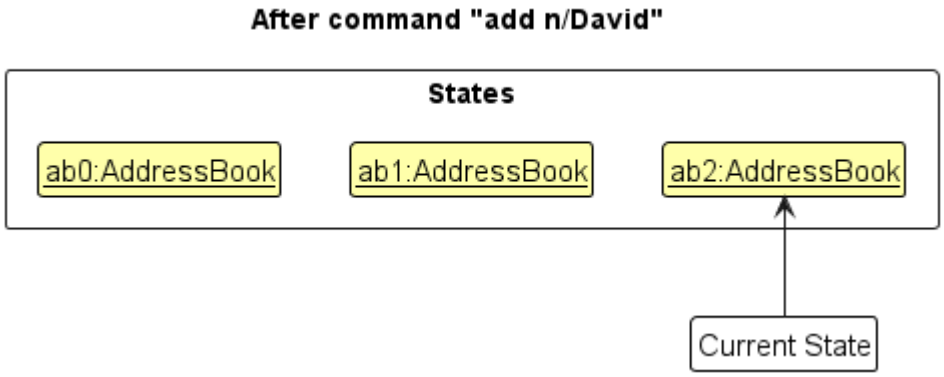
Step 1. The user launches the application for the first time. The `VersionedAddressBook` will be initialized with the initial address book state, and the `currentStatePointer` pointing to that single address book state.



Step 2. The user executes `delete 5` command to delete the 5th person in the address book. The `delete` command calls `Model#commitAddressBook()`, causing the modified state of the address book after the `delete 5` command executes to be saved in the `addressBookStateList`, and the `currentStatePointer` is shifted to the newly inserted address book state.

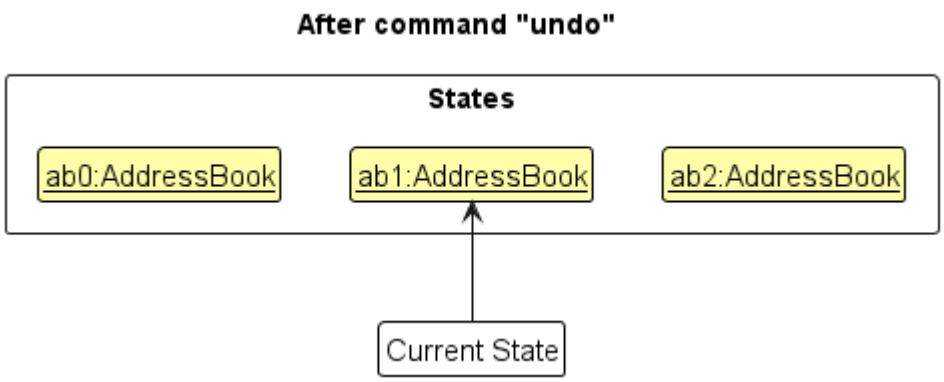


Step 3. The user executes `add n/David …` to add a new person. The `add` command also calls `Model#commitAddressBook()`, causing another modified address book state to be saved into the `addressBookStateList`.



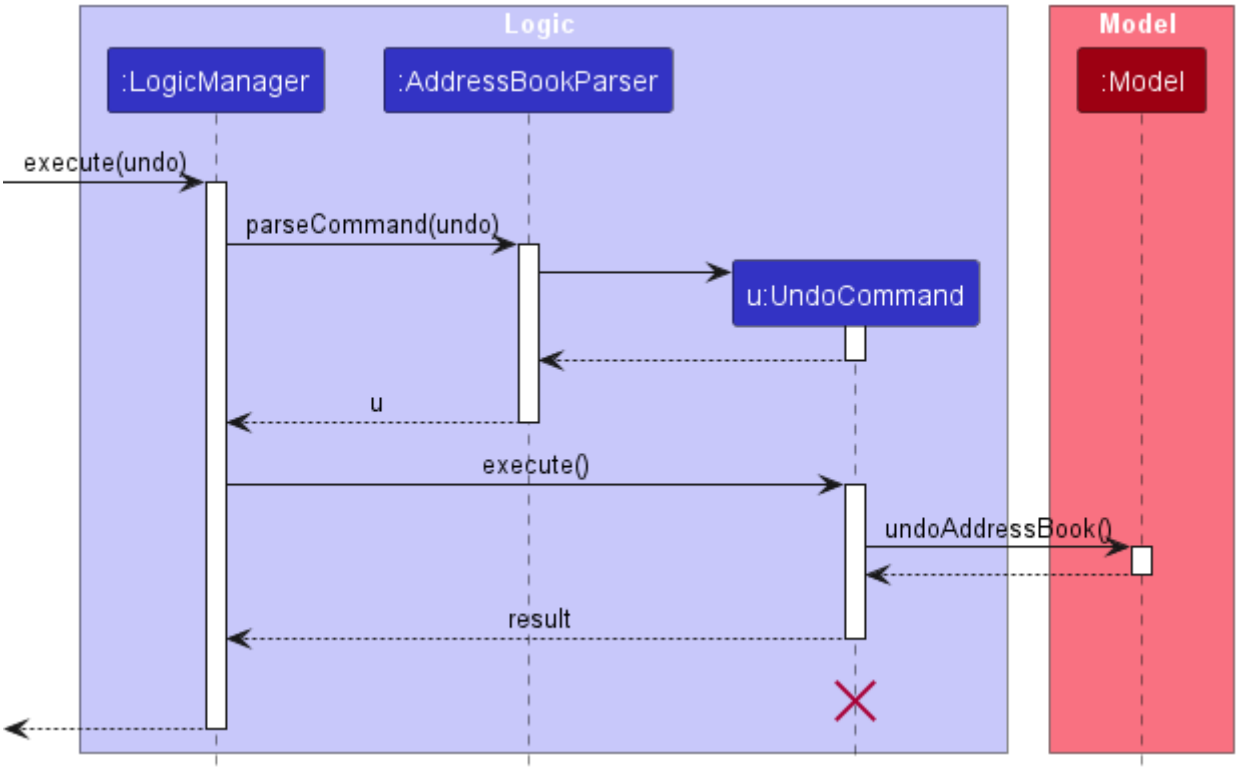
:information_source: **Note:** If a command fails its execution, it will not call `Model#commitAddressBook()`, so the address book state will not be saved into the `addressBookStateList`.

Step 4. The user now decides that adding the person was a mistake, and decides to undo that action by executing the `undo` command. The `undo` command will call `Model#undoAddressBook()`, which will shift the `currentStatePointer` once to the left, pointing it to the previous address book state, and restores the address book to that state.



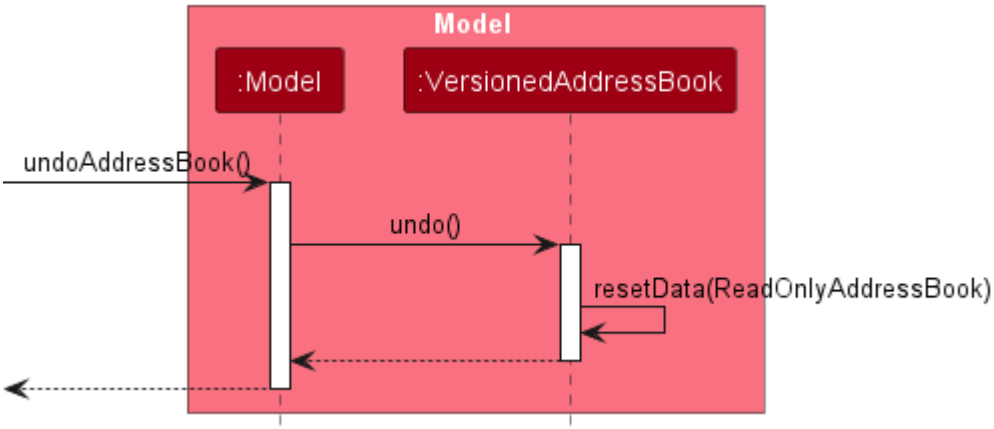
:information_source: **Note:** If the `currentStatePointer` is at index 0, pointing to the initial AddressBook state, then there are no previous AddressBook states to restore. The `undo` command uses `Model#canUndoAddressBook()` to check if this is the case. If so, it will return an error to the user rather than attempting to perform the undo.

The following sequence diagram shows how an undo operation goes through the `Logic` component:



:information_source: **Note:** The lifeline for `UndoCommand` should end at the destroy marker (X) but due to a limitation of PlantUML, the lifeline reaches the end of diagram.

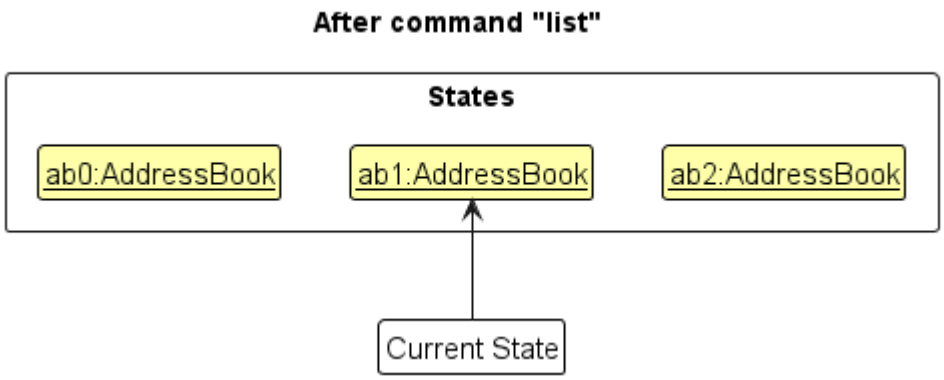
Similarly, how an undo operation goes through the `Model` component is shown below:



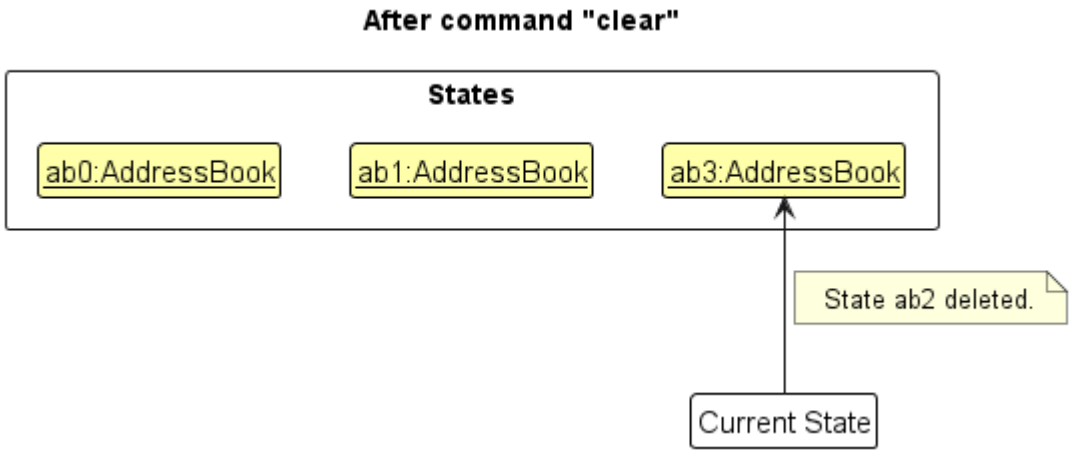
The `redo` command does the opposite—it calls `Model#redoAddressBook()`, which shifts the `currentStatePointer` once to the right, pointing to the previously undone state, and restores the address book to that state.

:information_source: **Note:** If the `currentStatePointer` is at index `addressBookStateList.size() - 1`, pointing to the latest address book state, then there are no undone AddressBook states to restore. The `redo` command uses `Model#canRedoAddressBook()` to check if this is the case. If so, it will return an error to the user rather than attempting to perform the redo.

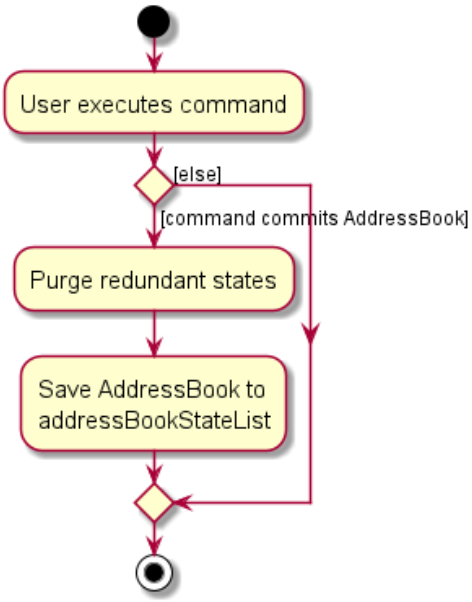
Step 5. The user then decides to execute the command `list`. Commands that do not modify the address book, such as `list`, will usually not call `Model#commitAddressBook()`, `Model#undoAddressBook()` or `Model#redoAddressBook()`. Thus, the `addressBookStateList` remains unchanged.



Step 6. The user executes `clear`, which calls `Model#commitAddressBook()`. Since the `currentStatePointer` is not pointing at the end of the `addressBookStateList`, all address book states after the `currentStatePointer` will be purged. Reason: It no longer makes sense to redo the `add n/David …` command. This is the behavior that most modern desktop applications follow.



The following activity diagram summarizes what happens when a user executes a new command:



Design considerations:

Aspect: How undo & redo executes:

- **Alternative 1 (current choice):** Saves the entire CrimeWatch data.
    - Pros: Easy to implement.
    - Cons: May have performance issues in terms of memory usage.
- **Alternative 2:** Individual command knows how to undo/redo by itself.
    - Pros: Will use less memory (e.g. for `delete`, just save the person being deleted).
    - Cons: We must ensure that the implementation of each individual command are correct.

{more aspects and alternatives to be added}

[Proposed] Data archiving

{Explain here how the data archiving feature will be implemented}

Documentation, logging, testing, configuration, dev-ops

- Documentation guide
- Testing guide
- Logging guide
- Configuration guide
- DevOps guide

Appendix: Requirements

Product scope

Target user profile:

- undercover law enforcement officer managing persons of interest
- comfortable with keyboard-first workflows
- needs fast access to contact context under time pressure
- needs encounter tracking and follow-up reminders

Value proposition: CrimeWatch helps officers manage and retrieve operational contact information quickly through a CLI-first workflow, while preserving encounter history and reminders.

User stories

Priorities: High (must have) - ∗ ∗ ∗, Medium (nice to have) - ∗ ∗, Low (unlikely to have) - ∗

Priority	As a …	I want to …	So that I can…
∗ ∗ ∗	undercover officer	create contact profiles	keep all suspect details organised in one secure place
∗ ∗ ∗	undercover officer	log encounters immediately after they happen	preserve accurate details while they are still fresh
∗ ∗ ∗	undercover officer	search contacts by name and tag	retrieve critical information quickly under time pressure
∗ ∗ ∗	undercover officer	update contact profiles	keep their information up to date
∗ ∗ ∗	undercover officer	delete contact profiles	remove any profile if no longer required
∗ ∗ ∗	undercover officer	record aliases and multiple identifiers for a contact	track individuals who use different identities
∗ ∗ ∗	undercover officer	add an email address to a contact	contact them via email if needed
∗ ∗	undercover officer	link contacts to each other	understand relationship networks within an investigation

Priority	As a …	I want to …	So that I can…
* *	undercover officer	tag contacts with statuses (active, inactive, high risk)	prioritise follow-ups effectively
* *	undercover officer	view a chronological timeline of interactions	understand the progression of a case at a glance
* *	undercover officer	attach notes and contextual observations to a contact	capture nuances that may not appear in formal reports
* *	undercover officer	quickly edit or update a contact's risk level	reflect changes in behaviour or threat level
* *	undercover officer	filter contacts by case or operation	focus only on relevant information
* *	undercover officer	log the location of each encounter	identify geographic patterns in suspect activity
* *	undercover officer	log outcomes of interactions	track whether objectives were achieved
* *	undercover officer with many contacts	group contacts (e.g. Case 1, Case 2)	organise my contacts more easily
*	undercover officer	mark follow-up reminders	ensure important leads are not forgotten
*	undercover officer	upload images and supporting documents of a contact	make relevant images and docs easily accessible
*	undercover officer	create a password to encrypt data on disk	ensure my data won't get stolen if my machine is compromised
*	undercover officer	view all encounter locations on a map	see how territories are related
*	undercover officer	export selected case information securely	prepare formal reports efficiently
*	undercover officer	view relationship maps between contacts	identify key influencers or central figures in a network
*	forgetful undercover officer	set a reminder time for a contact	remember to call or follow up at the right time

Use cases

(For all use cases below, the **System** is `CrimeWatch` and the **Actor** is the `officer`.)

Use case: Delete a person

MSS

1. User requests to list persons.
2. CrimeWatch shows a list of contacts.
3. User requests to delete a specific person in the list.
4. CrimeWatch deletes the contact.

   Use case ends.

Extensions

- 2a. The list is empty.

  Use case ends.

- 3a. The given index is invalid.

  - 3a1. CrimeWatch shows an error message.

    Use case resumes at step 2.

Use case: Edit a person

MSS

1. User requests to list persons.
2. CrimeWatch shows a list of contacts.
3. User requests to edit a specific person in the list with one or more fields.
4. CrimeWatch updates only the provided fields for that contact.

   Use case ends.

Extensions

- 2a. The list is empty.

  Use case ends.

- 3a. The given index is invalid.

  - 3a1. CrimeWatch shows an error message.

    Use case resumes at step 2.

- 3b. No editable field is provided.

  - 3b1. CrimeWatch shows an error message.

    Use case resumes at step 2.

Use case: Edit an encounter

MSS

1. User requests to view a contact profile.
2. CrimeWatch shows encounter cards with encounter indices.
3. User requests to edit a specific encounter using `PERSON_INDEX` and `ENCOUNTER_INDEX`.
4. CrimeWatch updates only the provided encounter fields.

   Use case ends.

Extensions

- 2a. The contact has no encounters.

  Use case ends.

- 3a. The given person index is invalid.
  - 3a1. CrimeWatch shows an error message.

    Use case resumes at step 2.

- 3b. The given encounter index is invalid.
  - 3b1. CrimeWatch shows an error message.

    Use case resumes at step 2.

- 3c. No editable field is provided.
  - 3c1. CrimeWatch shows an error message.

    Use case resumes at step 2.

Use case: View a protected contact

MSS

1. Officer requests to view a contact profile by index.
2. System detects contact is password-protected.
3. Officer supplies password.
4. System validates password and shows full profile.

Use case ends.

Extensions

- 1a. Index is invalid.
  - 1a1. System shows an error message. Use case ends.
- 3a. Password is missing or incorrect.
  - 3a1. System shows an error message. Use case ends.

Use case: Log an encounter

MSS

1. Officer enters `log` command with required encounter fields.
2. System validates input.
3. System appends encounter to target contact.
4. System confirms successful logging.

Use case ends.

Extensions

- 1a. Target index is invalid.
  - 1a1. System shows an error message.
- 2a. Required fields are missing/invalid.
  - 2a1. System shows an error message.

Non-Functional Requirements

1. Must run on mainstream OSes with Java `17` or higher.
2. Should support at least 1000 contacts with no noticeable lag for typical operations.
3. Common tasks should be faster via commands than equivalent mouse-heavy workflows.
4. Data file corruption and permission errors should fail gracefully with clear user feedback.
5. The design should remain maintainable under feature evolution (new command prefixes, additional `Person` fields).

Glossary

- **Contact**: person of interest tracked in CrimeWatch.
- **Stage**: investigation lifecycle marker (`surveillance`, `approached`, `cooperating`, `arrested`, `closed`).
- **Encounter**: time-stamped interaction record linked to a contact.
- **Reminder**: scheduled note linked to a contact.
- **Protected contact**: contact with a configured `pw/` value.

Appendix: Instructions for manual testing

Given below are manual tests for major functional paths and edge cases.

:information_source: These are baseline checks. Testers are expected to perform additional exploratory testing.

Launch and shutdown

1. Initial launch

    1. Download the jar file and place it in an empty folder.
    2. Run the jar file. Expected: app window opens with sample contacts.

2. Window preference persistence

    1. Resize and reposition the app window, then close the app.
    2. Relaunch the app. Expected: previous window size and location are restored.

Add and edit contact

1. Add contact with required + optional fields

    1. Test case: `add n/Mark Tan p/91234567 e/mark@example.com a/Blk 10 Clementi Ave 2 s/surveillance al/MT note/Seen near station r/high t/caseA`
    2. Expected: contact added with provided values.

2. Add contact with invalid stage

    1. Test case: `add n/Mark Tan p/91234567 e/mark@example.com a/Blk 10 Clementi Ave 2 s/unknown`
    2. Expected: command rejected with stage validation message.

3. Edit protected contact without password

    1. Prerequisite: target contact has password.
    2. Test case: `edit 1 n/New Name`
    3. Expected: command rejected; password required message shown.

Deleting a person

1. Deleting a person while all persons are being shown
    1. Prerequisites: List all persons using the `list` command. Multiple persons in the list.
    2. Test case: `delete 1` Expected: First contact is deleted from the list. Details of the deleted contact are shown in the status message. Timestamp in the status bar is updated.
    3. Test case: `delete 0` Expected: No person is deleted. Error details are shown in the status message. Status bar remains the same.
    4. Other incorrect delete commands to try: `delete`, `delete x`, `...` (where x is larger than the list size) Expected: Similar to previous.

View and password protection

1. View protected contact with correct password

    1. Prerequisite: contact at index 1 is protected with password `hunter2`.
    2. Test case: `view 1 pw/hunter2`
    3. Expected: full profile shown in view panel.

2. View protected contact with wrong password

    1. Test case: `view 1 pw/wrong`
    2. Expected: command rejected with incorrect password message.

Log encounter and edit encounter

1. Log encounter successfully

    1. Test case: `log 1 d/2026-04-09 t/18:30 l/Maxwell Road desc/Observed exchange out/No immediate action`
    2. Expected: encounter added and success message shown.

2. Edit encounter successfully

    1. Prerequisite: contact 1 has at least one encounter.
    2. Test case: `editencounter 1 1 out/Subject identified`
    3. Expected: first encounter for contact 1 updated.

Reminder and sort

1. Add reminder

    1. Test case: `remind 1 d/2026-05-01 t/09:00 note/Follow up with source`
    2. Expected: reminder added and shown in sorted order in profile view.

2. Sort by recent

    1. Test case: `sort recent`
    2. Expected: list reordered by latest encounter timestamp, most recent first.

Export and data persistence

1. Export by location

    1. Prerequisite: there are encounters at `Maxwell Road`.
    2. Test case: `export l/Maxwell Road`
    3. Expected: CSV generated under `exports/` with matching rows only.

2. Save/load smoke check

    1. Add or edit a contact.
    2. Exit and relaunch app.
    3. Expected: changes persist across restart.

3. Dealing with missing/corrupted data files

    1. Corrupt or remove the data file, then relaunch the app.
    2. Expected: the app handles the read failure gracefully and reports a clear error instead of crashing silently.